

Homework 6

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Math 38

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Problem 1. Show that hypercubes H_k have $\kappa(H_k) = k$.

Proof. We prove that $\kappa(H_k) = k$ by double enclosure.

Upper bound: H_k is k -regular, so $\kappa(H_k) \leq \delta(H_k) = k$ since for any graph $\kappa(G) \leq \delta(G)$.

Lower bound: We prove $\kappa(H_k) \geq k$ by induction on k .

Base case: $H_1 = K_2$, which has $\kappa(K_2) = 1$.

Inductive step: Assume $\kappa(H_{k-1}) \geq k-1$. Recall that H_k is constructed from two copies of H_{k-1} , call them Q_0 and Q_1 , with a perfect matching between corresponding vertices. Let S be a vertex cut of H_k with $|S| < k$; we derive a contradiction. Let $S_0 = S \cap V(Q_0)$ and $S_1 = S \cap V(Q_1)$. Since $|S| < k$, either $|S_0| < k-1$ or $|S_1| < k-1$ (or both). If $|S_0| < k-1$, then by the inductive hypothesis $Q_0 - S_0$ is connected. Similarly if $|S_1| < k-1$, then $Q_1 - S_1$ is connected.

Case 1: Both $|S_0| < k-1$ and $|S_1| < k-1$. Then $Q_0 - S_0$ and $Q_1 - S_1$ are both connected. Since $|S_0| < k-1$ and $Q_0 - S_0$ has at least $2^{k-1} - (k-2) \geq 2$ vertices, and each vertex in Q_0 has a neighbor in Q_1 via the matching, there exists a matching edge between $Q_0 - S_0$ and $Q_1 - S_1$. So $H_k - S$ is connected, contradicting that S is a cut.

Case 2: One side, say $|S_0| = k-1$ and $|S_1| \leq 0$, so $S_1 = \emptyset$. Then $Q_1 - S_1 = Q_1$ is connected. Every vertex in $Q_0 - S_0$ has a matching neighbor in Q_1 , so $Q_0 - S_0$ is connected to Q_1 through the matching edges. Thus $H_k - S$ is connected, which contradicts.

Therefore no vertex cut of size less than k exists, so $\kappa(H_k) \geq k$, proving $\kappa(H_k) = k$. $\square$

Problem 2. Let G be a connected graph with at least three vertices. Form G' from G by adding an edge with endpoints x, y whenever $d_G(x, y) = 2$. Prove that G' is 2-connected.

Proof. We show that G' has no cut-vertex, which for a connected graph on at least 3 vertices is equivalent to being 2-connected.

Let v be any vertex of G' . We show $G' - v$ is connected. Let a and b be any two vertices in $G' - v$. Since G is connected, there exists a path P from a to b in G . If P does not pass through v , then P exists in $G' - v$ and we are done.

If P passes through v , write $P = a \cdots u v w \cdots b$ where u and w are the neighbors of v on P . Since u and w are both adjacent to v in G , $d_G(u, w) \leq 2$. If $d_G(u, w) = 1$, then uw is the edge in G . If $d_G(u, w) = 2$, then uw is an edge in G' by construction. Regardless, $uw \in E(G')$. Therefore, we can replace the segment $u v w$ in P with the single edge uw , obtaining a walk from a to b in $G' - v$. Hence $G' - v$ is connected. Since v was arbitrary, G' has no cut-vertex, so G' is 2-connected. $\square$

Question 3. Let G be a weighted connected graph. The *cap* from an edge cut $[S, \bar{S}]$ is the maximum weight of its edges. The *value* of a spanning tree is the minimum weight of its edges. Prove that the maximum value of a spanning tree of G equals the minimum cap of an edge cut in G .

Proof. Let T^* be a spanning tree of maximum value, and let $\text{val}(T^*) = w^*$. Let $[S^*, \bar{S}^*]$ be an edge cut of minimum cap, and let $\text{cap}(S^*, \bar{S}^*) = c^*$.

Upper Bound: $w^* \leq c^*$. Any spanning tree T must contain at least one edge from every edge cut $[S, \bar{S}]$ (since T is connected). In particular, T^* contains some edge $e \in [S^*, \bar{S}^*]$. The value of T^* satisfies

$$w^* = \text{val}(T^*) \leq w(e) \leq \max(w(f)) = c^*.$$

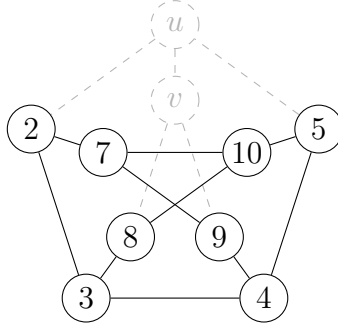
Lower Bound: $w^* \geq c^*$. Sort edges in decreasing order: $w(e_1) \geq w(e_2) \geq \dots \geq w(e_m)$. Build the max-value spanning tree T^* greedily: add e_i if it doesn't form a cycle. Let e^* be the last edge added (the minimum-weight edge of T^*), with $w(e^*) = w^*$.

At the moment before e^* is added, the forest has two components S and $\bar{S}$ that e^* connects. Every edge in $[S, \bar{S}]$ that was considered before e^* was skipped because it formed a cycle — but edges in $[S, \bar{S}]$ connect different components, so they cannot form a cycle. Therefore no edge in $[S, \bar{S}]$ was considered before e^* , meaning every edge in $[S, \bar{S}]$ has weight $\leq w(e^*) = w^*$. Thus $\text{cap}(S, \bar{S}) = \max(w(f)) \leq w^*$, so $c^* \leq w^*$. By double enclosure, $w^* = c^*$. $\square$

Question 4. Prove that the Petersen graph is 3-connected.

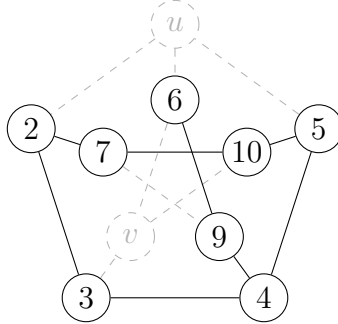
Proof. Since Petersen graph is 3-regular, it suffices to show $\kappa(P) \geq 3$, i.e., removing any two vertices leaves P connected. Let u and v be any two vertices of P . We show $P - \{u, v\}$ is connected.

Case 1: u and v are adjacent. Each has 3 neighbors, and they share no common neighbor (the Petersen graph has girth 5, so there is no triangle). Removing u and v removes at most $2 \times 3 - 1 = 5$ edges (the edge uv plus 2 other edges from each). The remaining graph $P - \{u, v\}$ has 8 vertices. Each of the 4 neighbors of $\{u, v\}$ loses one edge (to u or v), retaining degree 2. The other 4 vertices retain degree 3. Then, $P - \{u, v\}$ should be connected.



Case 1: u and v adjacent (edge $u-v$ removed)

Case 2: u and v are non-adjacent. Each has 3 neighbors. They share at most one common neighbor since the Petersen graph has girth 5). Removing u and v leaves 8 vertices. Again, $P - \{u, v\}$ remains connected.



Case 2: u and v non-adjacent (distance 2 apart)

Therefore $\kappa(P) \geq 3$, making it 3-connected. □

Question 5. Prove that if G is simple and $\delta(G) \geq n(G) - 2$, then $\kappa(G) = \delta(G)$. Is this the best possible?

Proof. Let $n = n(G)$ and $\delta = \delta(G) \geq n - 2$. Since $\delta \geq n - 2$, every vertex has degree either $n - 2$ or $n - 1$.

We always have $\kappa(G) \leq \delta(G)$, so it suffices to show $\kappa(G) \geq \delta$.

Case 1: $\delta = n - 1$. Then $G = K_n$, and $\kappa(K_n) = n - 1 = \delta$.

Case 2: $\delta = n - 2$. Every vertex has degree $n - 2$ or $n - 1$. A vertex of degree $n - 2$ is non-adjacent to exactly one other vertex.

Let S be a vertex cut of G with $|S| = \kappa(G)$. Then $G - S$ is disconnected. Let A and B be two components of $G - S$.

I claim that $|A| = |B| = 1$. Suppose $|A| \geq 2$. Let $a \in A$. Then a is adjacent to all of $A \setminus \{a\}$ and all of S (since the only vertices a might not be adjacent to are in B , and a misses at most one vertex overall). So $\deg(a) \geq (|A| - 1) + |S|$. But also $\deg(a) \leq n - 1$. Since a is not adjacent to any vertex in B , $\deg(a) = (|A| - 1) + |S| + 0 \leq n - 1$.

Meanwhile a has degree $\geq n - 2$, so $(|A| - 1) + |S| \geq n - 2$. Since $|A| + |B| + |S| = n$, we get $|A| - 1 + |S| = n - |B| - 1 \geq n - 2$, so $|B| \leq 1$.

Similarly, if $|B| \geq 2$, then $|A| \leq 1$. So at least one of A, B has size 1. Suppose $|B| = 1$, say $B = \{b\}$. Then b is adjacent to no vertex in A , so $\deg(b) = |S| = \kappa(G)$. Since $\deg(b) \geq \delta$, we get $\kappa(G) \geq \delta$.

Therefore $\kappa(G) = \delta(G)$. This is the best possible. There exist graphs with $\delta(G) = n - 3$ where $\kappa(G) < \delta(G)$. For example, take two copies of $K_{n/2}$ sharing a single vertex (for appropriate n). The graph has $\delta = n/2 - 1$ but $\kappa = 1$. More precisely, let $G = K_1 + (K_p \cup K_p)$ (a vertex joined to two disjoint cliques): then $\delta = p$ but $\kappa = 1$, demonstrating that without the condition $\delta \geq n - 2$, the equality can fail. $\square$